

2501.06662 — formula report

386 inline formulas, 60 display equations, 0 tables, 0 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00051	021	■ 0.536 ● C +5	$\begin{aligned} & \operatorname{Mag}(t \operatorname{\mathcal{M}}) \\ & = \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, y) \\ & = \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t. \end{aligned}$	$\operatorname{Mag}(t \operatorname{\mathcal{M}}) = \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$	$\operatorname{Mag}(t \operatorname{\mathcal{M}}) = \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$
2501.06662_E00059	023	■ 0.550 ● W +0	$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{\mathcal{H}(\operatorname{\mathcal{M}}, \pi(- \mid \perp) _{T(\perp)}, \zeta)}{n} \\ & = \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \log \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \end{aligned}$	$\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\operatorname{\mathcal{M}}, \pi(- \mid \perp) _{T(\perp)}, \zeta)}{n}$ $= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \log \pi(\perp a_1 \cdots a_{N-1} \mid \perp)$	$\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\operatorname{\mathcal{M}}, \pi(- \mid \perp) _{T(\perp)}, \zeta)}{n}$ $= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \log \pi(\perp a_1 \cdots a_{N-1} \mid \perp)$
2501.06662_E00058	023	■ 0.594 ● N +0	$\begin{aligned} & \mathcal{H}(\operatorname{\mathcal{M}}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\operatorname{\mathcal{M}})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t(y, z) p(z) \right) \\ & = - \sum_{y \in \operatorname{ob}(\operatorname{\mathcal{M}})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\operatorname{\mathcal{M}})} \pi(z \mid y)^t p(z) \right). \end{aligned}$	$\mathcal{H}(\operatorname{\mathcal{M}}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\operatorname{\mathcal{M}})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\operatorname{\mathcal{M}})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\operatorname{\mathcal{M}})} \pi(z \mid y)^t p(z) \right).$	$\mathcal{H}(\operatorname{\mathcal{M}}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\operatorname{\mathcal{M}})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\operatorname{\mathcal{M}})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\operatorname{\mathcal{M}})} \pi(z \mid y)^t p(z) \right).$
2501.06662_E00023	015	■ 0.636 ● N +2	$\begin{aligned} & \# \left\{ x \in \operatorname{ob}(\operatorname{\mathcal{L}}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1) = \left(\sum_{i=0}^{N-2} \# A^i \right) (\# A + 1) \\ & = \# A^{N-1} + 2 (\# A^{N-2}) + \cdots + 2 (\# A) + 1. \end{aligned}$	$\# \left\{ x \in \operatorname{ob}(\operatorname{\mathcal{L}}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1) = \left(\sum_{i=0}^{N-2} \# A^i \right) (\# A + 1)$ $= \# A^{N-1} + 2 (\# A^{N-2}) + \cdots + 2 (\# A) + 1$	$\# \left\{ x \in \operatorname{ob}(\operatorname{\mathcal{L}}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1) = \left(\sum_{i=0}^{N-2} \# A^i \right) (\# A + 1)$ $= \# A^{N-1} + 2 (\# A^{N-2}) + \cdots + 2 (\# A) + 1.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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2501.06662_EQ0052 (13)	021	0.693 N +0	$\operatorname{Mag}(t \operatorname{\mathcal{M}}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$		
2501.06662_EQ0033	017	0.720 W +2	$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathbf{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$		$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y x)^t & \text{if } y \in \mathbf{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise.} \end{cases}$
2501.06662_EQ0037 (10)	018	0.780 W +0	$\begin{aligned} \operatorname{Mag}(t \operatorname{\mathcal{M}}) &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} \sum_{y \in \mathbf{L}_x^{(1)}} \pi(y x)^t \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$		$\begin{aligned} \operatorname{Mag}(t \operatorname{\mathcal{M}}) &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} \sum_{y \in \mathbf{L}_x^{(1)}} \pi(y x)^t \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$
2501.06662_EQ0049	020	0.797 C +3	$\begin{aligned} \partial_k^i(y_0, \dots, y_k) &= \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases} \end{aligned}$		$\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$
2501.06662_EQ0001	003	0.822 N +0	$\operatorname{Mag}(t \operatorname{\mathcal{M}}) = (t-1) \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$		$\operatorname{Mag}(t \operatorname{\mathcal{M}}) = (t-1) \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$
2501.06662_EQ0029	016	0.829 W +1	$\left(\zeta_t - \delta \right)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i y_{i-1})^t$		$(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i y_{i-1})^t.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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2501.06662_EQ0036 (11)	018	■ 0.835 ● N +0	$\operatorname{Mag}(t \operatorname{\mathcal{M}}) = (t-1) \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$
2501.06662_EQ0026 (8)	015	■ 0.842 ● C +4	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$
2501.06662_EQ0021	014	■ 0.849 ● C +3	$\begin{aligned} \operatorname{Mag}(\operatorname{\mathsf{L}}) &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathsf{L}})} \zeta_{\operatorname{\mathsf{L}}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathsf{L}})} \mu_{\operatorname{\mathsf{L}}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathsf{L}})} \mu_{\operatorname{\mathsf{L}}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\operatorname{\mathsf{L}}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\operatorname{\mathsf{L}}}(x, x a_1) \\ &= \# \operatorname{ob}(\operatorname{\mathsf{L}}) - \# \left\{ x \in \operatorname{ob}(\operatorname{\mathsf{L}}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\operatorname{\mathsf{L}}) &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathsf{L}})} \zeta_{\operatorname{\mathsf{L}}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathsf{L}})} \mu_{\operatorname{\mathsf{L}}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathsf{L}})} \mu_{\operatorname{\mathsf{L}}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\operatorname{\mathsf{L}}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\operatorname{\mathsf{L}}}(x, x a_1) \\ &= \# \operatorname{ob}(\operatorname{\mathsf{L}}) - \# \left\{ x \in \operatorname{ob}(\operatorname{\mathsf{L}}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1) \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\operatorname{\mathsf{L}}) &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathsf{L}})} \zeta_{\operatorname{\mathsf{L}}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\operatorname{\mathsf{L}})} \mu_{\operatorname{\mathsf{L}}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\operatorname{\mathsf{L}})} \mu_{\operatorname{\mathsf{L}}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\operatorname{\mathsf{L}}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\operatorname{\mathsf{L}}}(x, x a_1) \\ &= \# \operatorname{ob}(\operatorname{\mathsf{L}}) - \# \left\{ x \in \operatorname{ob}(\operatorname{\mathsf{L}}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1). \end{aligned}$
2501.06662_EQ0022	015	■ 0.871 ● W +1	$\begin{aligned} \# \operatorname{ob}(\operatorname{\mathsf{L}}) &= 1 + \sum_{i=1}^{N-1} \# \operatorname{\mathsf{L}}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$	$\begin{aligned} \# \operatorname{ob}(\operatorname{\mathsf{L}}) &= 1 + \sum_{i=1}^{N-1} \# \operatorname{\mathsf{L}}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2 \end{aligned}$	$\begin{aligned} \# \operatorname{ob}(\operatorname{\mathsf{L}}) &= 1 + \sum_{i=1}^{N-1} \# \operatorname{\mathsf{L}}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$
2501.06662_EQ0045	019	■ 0.922 ● N +1	$\zeta_{\operatorname{\mathcal{M}}}^{-1} = \zeta_{\operatorname{\mathcal{M}}}^{-1} \big _{\operatorname{ob}(\operatorname{\mathcal{M}}_x) \times \operatorname{ob}(\operatorname{\mathcal{M}}_x)},$	$\zeta_{\operatorname{\mathcal{M}}_x}^{-1} = \zeta_{\operatorname{\mathcal{M}}}^{-1} \big _{\operatorname{ob}(\operatorname{\mathcal{M}}_x) \times \operatorname{ob}(\operatorname{\mathcal{M}}_x)},$	$\zeta_{\operatorname{\mathcal{M}}_x}^{-1} = \zeta_{\operatorname{\mathcal{M}}}^{-1} \big _{\operatorname{ob}(\operatorname{\mathcal{M}}_x) \times \operatorname{ob}(\operatorname{\mathcal{M}}_x)},$
2501.06662_EQ0039 (12)	018	■ 0.940 ● N +0	$f'(1) = \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} H(p_x).$	$f'(1) = \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} H(p_x).$	$f'(1) = \sum_{x \in \operatorname{ob}(\operatorname{\mathcal{M}}) \setminus T(\perp)} H(p_x).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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2501.06662_E00046	020	■ 0.942 ● W +0	$\begin{aligned} & \operatorname{Mag}(\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ & = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ & = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)). \end{aligned}$	$\operatorname{Mag}(t\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$	$\operatorname{Mag}(t\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$
2501.06662_E00040	019	■ 0.945 ● K +0	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$
2501.06662_E00044	019	■ 0.971 ● K +0	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$
2501.06662_E00032	017	■ 0.973 ● C +4	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$
2501.06662_E00056	022	■ 0.986 ● N +0	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$
2501.06662_E00015 (7)	012	■ 0.988 ● N +0	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$
2501.06662_E00002	004	■ 0.991 ● N +0	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$
2501.06662_E00005	006	■ 0.991 ● N +1	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$
2501.06662_E00006	008	■ 0.996 ● W +0	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } y \leq N \}.$	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } y \leq N \}.$	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } y \leq N \}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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2501.06662_E00034	017	0.998 N +0	$\operatorname{Mag}(t \mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$	$\operatorname{Mag}(t \mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$	$\operatorname{Mag}(t \mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$
2501.06662_E00003	004	0.998 C +6	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$
2501.06662_E00054 (14)	022	0.998 N +0	$\operatorname{Mag}(t \mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$	$\operatorname{Mag}(t \mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$	$\operatorname{Mag}(t \mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$
2501.06662_E00019	014	0.999 N +2	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$
2501.06662_E00042	019	0.999 N +0	$H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1},$	$H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1},$	$H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1},$
2501.06662_E00004	005	1.000 N +0	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$
2501.06662_E00012	011	1.000 N +1	$\begin{aligned} & x = \perp a_1 \cdots a_t \\ & y = \perp a_1 \cdots a_t \cdots a_{t+k} \\ & z = \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$	$\begin{aligned} & x = \perp a_1 \cdots a_t \\ & y = \perp a_1 \cdots a_t \cdots a_{t+k} \\ & z = \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$	$\begin{aligned} & x = \perp a_1 \cdots a_t \\ & y = \perp a_1 \cdots a_t \cdots a_{t+k} \\ & z = \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'}. \end{aligned}$
2501.06662_E00031	017	1.000 W +1	$\prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$	$\prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$	$\prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00007	008	■ 1.000 ● K +0	$\mathrm{p}\left(\mathrm{a}_{\mathrm{3}}\mid\mathrm{perp}\mathrm{a}\right)\mathrm{p}\left(\mathrm{a}_{\mathrm{4}}\mid\mathrm{perp}\mathrm{a}\mathrm{a}_{\mathrm{3}}\right)\mathrm{p}\left(\mathrm{a}_{\mathrm{5}}\mid\mathrm{perp}\mathrm{a}\mathrm{a}_{\mathrm{3}}\mathrm{a}_{\mathrm{4}}\right)\mathrm{p}\left(\dagger\mid\mathrm{perp}\mathrm{a}\mathrm{a}_{\mathrm{3}}\mathrm{a}_{\mathrm{4}}\mathrm{a}_{\mathrm{5}}\right).$	$p\left(a_3\mid a\right)p\left(a_4\mid aa_3\right)p\left(a_5\mid aa_3a_4\right)p\left(\dagger\mid aa_3a_4a_5\right).$	$p(a_3 \perp a)p(a_4 \perp aa_3)p(a_5 \perp aa_3a_4)p(\dagger \perp aa_3a_4a_5).$
2501.06662_E00008 (1)	009	■ 1.000 ● W +0	$\mathrm{\pi}(y\mid x):=\left\{\begin{array}{l}1\text{ \& \text{if } x=y}\\0\text{ \& \text{if } x\nrightarrow y}\\\prod_{i=1}^kp\left(a_{t+i}\mid y_{<t+i}\right)\text{ \& \text{if } x\rightarrow y}\end{array}\right.$	$\pi(y\mid x):=\left\{\begin{array}{l}1\text{ \& \text{if } x=y}\\0\text{ \& \text{if } x\nrightarrow y}\\\prod_{i=1}^kp\left(a_{t+i}\mid y_{<t+i}\right)\text{ \& \text{if } x\rightarrow y}\end{array}\right.$	$\pi(y x):=\left\{\begin{array}{l}1\text{ \& \text{if } x=y}\\0\text{ \& \text{if } x\nrightarrow y}\\\prod_{i=1}^kp(a_{t+i} y_{<t+i})\text{ \& \text{if } x\rightarrow y}\end{array}\right.$
2501.06662_E00009 (2)	009	■ 1.000 ● N +0	$\sum_{y\in T(x)}\mathrm{\pi}(y\mid x)=\sum_{a\in A\cup\{\dagger\}}p_x(a)=1,$	$\sum_{y\in T(x)}\pi(y\mid x)=\sum_{a\in A\cup\{\dagger\}}p_x(a)=1,$	$\sum_{y\in T(x)}\pi(y x)=\sum_{a\in A\cup\{\dagger\}}p_x(a)=1,$
2501.06662_E00010 (4)	009	■ 1.000 ● W +0	$\begin{aligned}\sum_{y\in T(x)}\mathrm{\pi}(y\mid x)\text{ \& }&=\sum_{i=0}^{m-1}\sum_{a^{\prime}\in A^i}\mathrm{\pi}\left(xa^{\prime}\dagger\mid x\right)+\sum_{a\in A^m}\mathrm{\pi}\left(xa\mid x\right)\\&=\sum_{i=0}^{m-1}\sum_{a^{\prime}\in A^i}\mathrm{\pi}\left(xa^{\prime}\dagger\mid x\right)+\sum_{\substack{a=a^{\prime}a^{\prime\prime}\\a^{\prime}\in A^{m-1},a^{\prime\prime}\in A}}p\left(a^{\prime\prime}\mid xa^{\prime}\right)\mathrm{\pi}\left(xa^{\prime}\mid x\right)\\&=\sum_{i=0}^{m-2}\sum_{a^{\prime}\in A^i}\mathrm{\pi}\left(xa^{\prime}\dagger\mid x\right)+\sum_{a^{\prime}\in A^{m-1}}\mathrm{\pi}\left(xa^{\prime}\mid x\right)\sum_{a^{\prime\prime}\in A\cup\{\dagger\}}p\left(a^{\prime\prime}\mid xa^{\prime}\right)\\&=\sum_{i=0}^{m-2}\sum_{a^{\prime}\in A^i}\mathrm{\pi}\left(xa^{\prime}\dagger\mid x\right)+\sum_{a^{\prime}\in A^{m-1}}\mathrm{\pi}\left(xa^{\prime}\mid x\right)\end{aligned}$	$\begin{aligned}\sum_{y\in T(x)}\pi(y\mid x)&=\sum_{i=0}^{m-1}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{a\in A^m}\pi(xa\mid x)\\&=\sum_{i=0}^{m-1}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{\substack{a=a^{\prime}a^{\prime\prime}\\a^{\prime}\in A^{m-1},a^{\prime\prime}\in A}}p(a^{\prime\prime}\mid xa^{\prime})\pi(xa^{\prime}\mid x)\\&=\sum_{i=0}^{m-2}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{a^{\prime}\in A^{m-1}}\pi(xa^{\prime}\mid x)\sum_{a^{\prime\prime}\in A\cup\{\dagger\}}p(a^{\prime\prime}\mid xa^{\prime})\\&=\sum_{i=0}^{m-2}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{a^{\prime}\in A^{m-1}}\pi(xa^{\prime}\mid x)\end{aligned}$	$\begin{aligned}\sum_{y\in T(x)}\pi(y x)&=\sum_{i=0}^{m-1}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{a\in A^m}\pi(xa\mid x)\\&=\sum_{i=0}^{m-1}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{\substack{a=a^{\prime}a^{\prime\prime}\\a^{\prime}\in A^{m-1},a^{\prime\prime}\in A}}p(a^{\prime\prime} xa^{\prime})\pi(xa^{\prime} x)\\&=\sum_{i=0}^{m-2}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{a^{\prime}\in A^{m-1}}\pi(xa^{\prime} x)\sum_{a^{\prime\prime}\in A\cup\{\dagger\}}p(a^{\prime\prime} xa^{\prime})\\&=\sum_{i=0}^{m-2}\sum_{a^{\prime}\in A^i}\pi(xa^{\prime}\dagger\mid x)+\sum_{a^{\prime}\in A^{m-1}}\pi(xa^{\prime} x)\end{aligned}$
2501.06662_E00011	010	■ 1.000 ● N +0	$\begin{aligned}1\text{ \& \text{\&leq }\mathcal{C}(x,x)}\\&\text{ \& \text{\&leq }\mathcal{C}(y,z)\otimes\mathcal{C}(x,y)\text{ \& \text{\&leq }\mathcal{C}(x,z)}}\end{aligned}$	$\begin{aligned}1\text{ \& \text{\&leq }\mathcal{C}(x,x)}\\&\mathcal{C}(y,z)\otimes\mathcal{C}(x,y)\text{ \& \text{\&leq }\mathcal{C}(x,z)}\end{aligned}$	$\begin{aligned}1\text{ \& \text{\&leq }\mathcal{C}(x,x)}\\&\mathcal{C}(y,z)\otimes\mathcal{C}(x,y)\text{ \& \text{\&leq }\mathcal{C}(x,z)}\end{aligned}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00013 (6)	011	■ 1.000 ● N +1	$\begin{aligned} \pi(y \mid x) \pi(z \mid y) &= \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j}) \\ &= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i}) \\ &= \pi(z \mid x). \end{aligned}$	$\pi(y \mid x) \pi(z \mid y) = \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i})$ $= \pi(z \mid x).$	$\pi(y x)\pi(z y) = \prod_{i=1}^k p(a_{t+i} y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} z_{<t+i})$ $= \pi(z x).$
2501.06662_E00014	012	■ 1.000 ● K +0	$d(x, y) := -\ln \pi(y \mid x) \ .$	$d(x,y) := -\ln \pi(y \mid x).$	$d(x,y) := -\ln \pi(y x).$
2501.06662_E00016	013	■ 1.000 ● N +0	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$
2501.06662_E00017	013	■ 1.000 ● N +0	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$
2501.06662_E00018	013	■ 1.000 ● N +1	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$
2501.06662_E00020	014	■ 1.000 ● W +0	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$
2501.06662_E00024	015	■ 1.000 ● K +0	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y x),$
2501.06662_E00025	015	■ 1.000 ● N +0	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t.$	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t.$	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y x)^t.$
2501.06662_E00027 (9)	016	■ 1.000 ● N +1	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k,$
2501.06662_E00028	016	■ 1.000 ● N +0	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$
2501.06662_E00030	016	■ 1.000 ● K +0	$\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$	$\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$	$\zeta_t^{-1}(x, y) = \pi(y x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00035	017	■ 1.000 ● N +0	$H_{\{t\}}(p)=\frac{1}{t-1}\left(1-\sum_{i=1}^n p_{\{i\}}^{\{t\}}\right)$.	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right).$	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right).$
2501.06662_E00038	018	■ 1.000 ● C +3	$\lim_{t \rightarrow 1} H_{\{t\}}(p)=\lim_{t \rightarrow 1} \frac{1}{t-1}\left(1-\sum_{s \in S} p(s)^{\{t\}}\right)=-\sum_{s \in S} p(s) \ln p(s)=: H(p)$.	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$
2501.06662_E00041	019	■ 1.000 ● K +0	$Z_{\{x\}}(t)=\sum_{a \in A \cup \{\dagger\}} p_{\{x\}}(a)^{\{t\}}=\sum_{a \in A \cup \{\dagger\}} e^{\{-t\left(-\ln p_{\{x\}}(a)\right)\}}$	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$
2501.06662_E00043	019	■ 1.000 ● N +0	$\mathbb{E}_{\rho}(E)=\sum_{s \in S} E(s) \frac{e^{\{-t E(s)\}}}{\tilde{Z}(t)}=-\frac{\mathrm{d}}{\mathrm{d} t} \ln \tilde{Z}(t)=\frac{f'(t)}{\tilde{Z}(t)},$ $\ln \tilde{Z}(t)=\frac{f(\mathrm{prime}(t))}{\tilde{Z}(t)},$	$\mathbb{E}_{\rho}(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{\mathrm{d}}{\mathrm{d} t} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$	$\mathbb{E}_{\rho}(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{\mathrm{d}}{\mathrm{d} t} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$
2501.06662_E00047	020	■ 1.000 ● W +2	$G_{\{k, \ell\}}=\left(\left(y_{\{0\}}, \ldots, y_{\{k\}}\right) \in M^{\{k+1\}} \mid \sum_{i=0}^{k-1} d\left(y_{\{i\}}, y_{\{i+1\}}\right)=\ell \text { and for all } i, y_{\{i\}} \neq y_{\{i+1\}}\right)$.	$G_{k, \ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$	$G_{k, \ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$
2501.06662_E00048	020	■ 1.000 ● K +0	$\partial_{\{k\}}=\sum_{i=0}^k(-1)^{\{i\}} \partial_{\{k\}}^{\{i\}}$	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$
2501.06662_E00050	021	■ 1.000 ● N +0	$\operatorname{Mag}(t \mathcal{M})=\sum_{\ell} q^{\ell} \sum_{k \geq 0}(-1)^{\{k\}} \operatorname{rank}\left(H_{\{k, \ell\}}(\mathcal{M})\right)$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} q^{\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank} \left(H_{k, \ell}(\mathcal{M}) \right)$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} q^{\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank} (H_{k, \ell}(\mathcal{M}))$
2501.06662_E00053	021	■ 1.000 ● N +0	$\sum_{c \in G_{\{k, \ell\}}}(-1)^{\{k\}} e^{\{-t \ell\}}=(-1)^{\{k\}} q^{\{\ell\}} \# G_{\{k, \ell\}}=\sum_{c \in G_{k, \ell}}(-1)^k q^{\ell} \operatorname{rank}\left(M C_{\{k, \ell\}}(\mathcal{M})\right)$.	$\sum_{c \in G_{k, \ell}} (-1)^k e^{-t \ell} = (-1)^k q^{\ell} \# G_{k, \ell} = (-1)^k q^{\ell} \operatorname{rank} (M C_{k, \ell}(\mathcal{M})).$	$\sum_{c \in G_{k, \ell}} (-1)^k e^{-t \ell} = (-1)^k q^{\ell} \# G_{k, \ell} = (-1)^k q^{\ell} \operatorname{rank} (M C_{k, \ell}(\mathcal{M})).$
2501.06662_E00055	022	■ 1.000 ● N +0	$P P L(x)=\exp \left\{-\frac{1}{n} \sum_{i=1}^n \ln p\left(a_{\{i\}} \mid y_{\{<i\}}\right)\right\}$	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i \mid y_{<i}) \right\}$	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i \mid y_{<i}) \right\}$
2501.06662_E00057	023	■ 1.000 ● K +0	$\theta(a, b)=0 \quad \text { if } \quad \operatorname{A}(a, b)=\varnothing$.	$\theta(a, b) = 0 \quad \text{if} \quad A(a, b) = \varnothing.$	$\theta(a, b) = 0 \quad \text{if} \quad A(a, b) = \varnothing.$
2501.06662_E00060	023	■ 1.000 ● N +0	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P\left(a, a'\right) \log P\left(a, a'\right),$	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00001	004	—	t	t
2501.06662_F00002	001	—	$\pi(y \mid x) \in [0,1]$	$\pi(y \mid x) \in [0,1]$
2501.06662_F00003	001	—	x	x
2501.06662_F00004	001	—	y	y
2501.06662_F00005	002	—	$\pi(y \mid x)$	$\pi(y \mid x)$
2501.06662_F00006	002	—	n	n
2501.06662_F00007	002	—	(n-1)	$(n-1)$
2501.06662_F00008	002	—	$\dagger$	$\dagger$
2501.06662_F00009	002	—	a	a
2501.06662_F00010	002	—	$x=\perp a$	$x=\perp a$
2501.06662_F00011	002	—	p_x	p_x
2501.06662_F00012	002	—	a_1	a_1
2501.06662_F00013	002	—	$x\,a_1$	xa_1
2501.06662_F00014	002	—	p_{xa}	p_{xa}
2501.06662_F00015	002	—	a_2	a_2
2501.06662_F00016	002	—	$a_2=\dagger$	$a_2=\dagger$
2501.06662_F00017	002	—	aa_1	aa_1
2501.06662_F00018	002	—	π	π
2501.06662_F00019	003	—	$\pi(x \mid x)=1$	$\pi(x \mid x)=1$
2501.06662_F00020	003	—	$\pi(- \mid x)$	$\pi(- \mid x)$
2501.06662_F00021	003	—	$T(x)$	$T(x)$
2501.06662_F00022	003	—	$[0,\infty]$	$[0,\infty]$
2501.06662_F00023	003	—	$-\ln \pi(y \mid x)$	$-\ln \pi(y \mid x)$
2501.06662_F00024	003	—	$([0,\infty],\geq,+,\emptyset)$	$([0,\infty],\geq,+,\emptyset)$
2501.06662_F00025	003	—	$\mathcal{M}$	$\mathcal{M}$
2501.06662_F00026	003	—	$x,y,\ldots$	$x,y,\ldots$
2501.06662_F00027	003	—	$d(x,y)=-\ln \pi(y \mid x)$	$d(x,y)=-\ln \pi(y \mid x)$
2501.06662_F00028	003	—	$\operatorname{Mag}(t\,\mathcal{M})$	$\operatorname{Mag}(t\mathcal{M})$
2501.06662_F00029	003	—	$t>0$	$t>0$
2501.06662_F00030	004	—	$\#(T(\perp))$	$\#(T(\perp))$
2501.06662_F00031	004	—	H_t	H_t
2501.06662_F00032	004	—	$H_t\left(p_1,\ldots,p_m\right)=\left(1-\sum_{i=1}^m p_i^t\right) /\left(t-1\right)$	$H_t\left(p_1,\ldots,p_m\right)=\left(1-\sum_{i=1}^m p_i^t\right) /\left(t-1\right)$
2501.06662_F00033	004	—	$\left(H_t\right)_{t \in\left(0, \infty\right) \backslash\{1\}}$	$\left(H_t\right)_{t \in\left(0, \infty\right) \backslash\{1\}}$
2501.06662_F00034	004	—	$t>1$	$t>1$
2501.06662_F00035	004	—	$H_t\left(p_x\right)$	$H_t\left(p_x\right)$
2501.06662_F00036	004	—	$\left(n^{1-t}-1\right) /\left(1-t\right)$	$\left(n^{1-t}-1\right) /\left(1-t\right)$
2501.06662_F00037	004	—	$t \rightarrow \infty$	$t \rightarrow \infty$
2501.06662_F00038	004	—	$\mathcal{M}_1$	$\mathcal{M}_1$
2501.06662_F00039	004	—	$\mathcal{M}_2$	$\mathcal{M}_2$
2501.06662_F00040	004	—	$p_{\bullet}^{(1)}$	$p_{\bullet}^{(1)}$
2501.06662_F00041	004	—	$p_{\bullet}^{(2)}$	$p_{\bullet}^{(2)}$
2501.06662_F00042	004	—	x'	x'
2501.06662_F00043	004	—	$H_t\left(p_{x'}^{(1)}\right)>H_t\left(p_x^{(2)}\right)$	$H_t\left(p_{x'}^{(1)}\right)>H_t\left(p_x^{(2)}\right)$
2501.06662_F00044	004	—	$\operatorname{Mag}\left(t\,\mathcal{M}_1\right)>\operatorname{Mag}\left(t\,\mathcal{M}_2\right)$	$\operatorname{Mag}\left(t\mathcal{M}_1\right)>\operatorname{Mag}\left(t\mathcal{M}_2\right)$
2501.06662_F00045	004	—	$t<1$	$t<1$
2501.06662_F00046	004	—	$\lim _{t \rightarrow 0} \operatorname{Mag}(t\,\mathcal{M})=(1-n) \cdot \#(\operatorname{ob}(\mathcal{M}) \backslash T(\perp))+\#(T(\perp))$	$\lim _{t \rightarrow 0} \operatorname{Mag}(t\mathcal{M})=(1-n) \cdot \#(\operatorname{ob}(\mathcal{M}) \backslash T(\perp))+\#(T(\perp))$
2501.06662_F00047	004	—	$n>1$	$n>1$
2501.06662_F00048	004	—	$t=1$	$t=1$
2501.06662_F00049	004	—	$\sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H\left(p_x\right)$	$\sum_{x \in \operatorname{ob}(\mathcal{M}) \backslash T(\perp)} H\left(p_x\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00050	004	—	$\mathcal{M}_x$	$\mathcal{M}_x$
2501.06662_F00051	004	—	$\left(\sum_y \mu(x, y)\right)_x$	$\left(\sum_y \mu(x, y)\right)_x$
2501.06662_F00052	005	—	μ	μ
2501.06662_F00053	005	—	ℓ	ℓ
2501.06662_F00054	005	—	$\mathcal{L}$	$\mathcal{L}$
2501.06662_F00055	005	—	$-\ln : [0, 1] \rightarrow [0, \infty]$	$-\ln : [0, 1] \rightarrow [0, \infty]$
2501.06662_F00056	006	—	A	A
2501.06662_F00057	006	—	$a, b, \ldots$	$a, b, \dots$
2501.06662_F00058	006	—	$A^{\ast}$	$A^{\ast}$
2501.06662_F00059	006	—	$a \leq b$	$a \leq b$
2501.06662_F00060	006	—	b	b
2501.06662_F00061	006	—	$b = a a^{\prime}$	$b = a a^{\prime}$
2501.06662_F00062	006	—	$a^{\prime} \in A^{\ast}$	$a^{\prime} \in A^{\ast}$
2501.06662_F00063	006	—	$\leq$	$\leq$
2501.06662_F00064	006	—	ϵ	ϵ
2501.06662_F00065	006	—	$ a $	$ a $
2501.06662_F00066	006	—	$a \in A^{\ast}$	$a \in A^{\ast}$
2501.06662_F00067	006	—	$(P, \leq)$	$(P, \leq)$
2501.06662_F00068	006	—	P	P
2501.06662_F00069	006	—	$x \rightarrow y$	$x \rightarrow y$
2501.06662_F00070	006	—	$x \leq y$	$x \leq y$
2501.06662_F00071	006	—	$N \in \mathbb{N}$	$N \in \mathbb{N}$
2501.06662_F00072	006	—	$\mathrm{L} := \mathrm{L}^{\leq N}$	$\mathrm{L} := \mathrm{L}^{\leq N}$
2501.06662_F00073	006	—	$\left((A \cup \{\perp, \dagger\})^{\ast}, \leq\right)$	$((A \cup \{\perp, \dagger\})^{\ast}, \leq)$
2501.06662_F00074	006	—	$N-1$	$N-1$
2501.06662_F00075	006	—	x, y	x, y
2501.06662_F00076	006	—	$(A \cup \{\perp, \dagger\})^{\ast}$	$(A \cup \{\perp, \dagger\})^{\ast}$
2501.06662_F00077	006	—	$\perp a$	$\perp a$
2501.06662_F00078	006	—	$\perp a \dagger$	$\perp a \dagger$
2501.06662_F00079	006	—	$ x $	$ x $
2501.06662_F00080	006	—	$ \perp = 1$	$ \perp = 1$
2501.06662_F00081	006	—	$\operatorname{ob}(\mathrm{L})$	$\operatorname{ob}(\mathrm{L})$
2501.06662_F00082	006	—	$\bigsqcup_{j=0}^{N-1} \mathrm{L}^{\sim(j)}$	$\bigsqcup_{j=0}^{N-1} \mathrm{L}^{(j)}$
2501.06662_F00083	006	—	$\mathrm{L}^{\sim(j)}$	$\mathrm{L}^{(j)}$
2501.06662_F00084	006	—	$1+j$	$1+j$
2501.06662_F00085	006	—	j	j
2501.06662_F00086	006	—	$\mathrm{L}^{\sim(j)}$	$\mathrm{L}^{(j)}$
2501.06662_F00087	006	—	$\mathrm{L}^{\sim(i)}$	$\mathrm{L}^{(i)}$
2501.06662_F00088	006	—	$j > i$	$j > i$
2501.06662_F00089	006	—	L_x	L_x
2501.06662_F00090	006	—	$y \in \mathrm{L}$	$y \in \mathrm{L}$
2501.06662_F00091	006	—	$\operatorname{ob}(\mathrm{L}_x) = \bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{(j)}$	$\operatorname{ob}(\mathrm{L}_x) = \bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{(j)}$
2501.06662_F00092	006	—	$\mathrm{L}_x^{(j)}$	$\mathrm{L}_x^{(j)}$
2501.06662_F00093	006	—	$ x +j$	$ x +j$
2501.06662_F00094	006	—	$\mathrm{L}_x^{(0)} = \{x\}$	$\mathrm{L}_x^{(0)} = \{x\}$
2501.06662_F00095	006	—	$\mathrm{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$	$\mathrm{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$
2501.06662_F00096	006	—	$\mathrm{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$	$\mathrm{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$
2501.06662_F00097	006	—	$\mathrm{L}_{\perp}^{(j)} = \mathrm{L}^{(j)}$	$\mathrm{L}_{\perp}^{(j)} = \mathrm{L}^{(j)}$
2501.06662_F00098	006	—	$j \geq 0$	$j \geq 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00099		—	$\top$	$\top$
2501.06662_F00100	009	—	$\mathbf{L}$	$\mathbf{L}$
2501.06662_F00101	007	—	$p_{\{x\}}:=p(-\mid x):A\cup\{\dagger\}\rightarrow[0,1]$	$p_x:=p(-\mid x):A\cup\{\dagger\}\rightarrow[0,1]$
2501.06662_F00102	007	—	$a_1=\dagger$	$a_1=\dagger$
2501.06662_F00103	007	—	$p_{\perp aa_1}$	$p_{\perp aa_1}$
2501.06662_F00104	007	—	$A\cup\{\dagger\}$	$A\cup\{\dagger\}$
2501.06662_F00105	007	—	$p_{\perp aa_1 a_2}$	$p_{\perp aa_1 a_2}$
2501.06662_F00106	007	—	$y=x a_1\cdots a_{N-\mid x\mid}$	$y=x a_1\cdots a_{N-\mid x\mid}$
2501.06662_F00107	007	—	N	N
2501.06662_F00108	007	—	$a=a(x)\in A\cup\{\dagger\}$	$a=a(x)\in A\cup\{\dagger\}$
2501.06662_F00109	007	—	$p_x(a)=1$	$p_x(a)=1$
2501.06662_F00110	007	—	$p_x(a')=0$	$p_x(a')=0$
2501.06662_F00111	007	—	$a'\in(A\cup\{\dagger\})\setminus\{a\}$	$a'\in(A\cup\{\dagger\})\setminus\{a\}$
2501.06662_F00112	007	—	$p_{\perp a_1\cdots a_n}(-)$	$p_{\perp a_1\cdots a_n}(-)$
2501.06662_F00113	007	—	a_n	a_n
2501.06662_F00114	007	—	$n\geq 1$	$n\geq 1$
2501.06662_F00115	007	—	$\tilde{A}=A\cup\{\dagger\}$	$\tilde{A}=A\cup\{\dagger\}$
2501.06662_F00116	007	—	$P_n(a_n,a_{n+1}):=p_{\perp a_1\cdots a_n}(a_{n+1})$	$P_n(a_n,a_{n+1}):=p_{\perp a_1\cdots a_n}(a_{n+1})$
2501.06662_F00117	007	—	$P_n:\tilde{A}\times\tilde{A}\rightarrow[0,1]$	$P_n:\tilde{A}\times\tilde{A}\rightarrow[0,1]$
2501.06662_F00118	007	—	$\sum_{a'\in\tilde{A}}P_n(a,a')=1$	$\sum_{a'\in\tilde{A}}P_n(a,a')=1$
2501.06662_F00119	007	—	$P_n(\dagger,\dagger)=1$	$P_n(\dagger,\dagger)=1$
2501.06662_F00120	007	—	$P_n(\dagger,a)=0$	$P_n(\dagger,a)=0$
2501.06662_F00121	007	—	$a\in A$	$a\in A$
2501.06662_F00122	007	—	$n>2$	$n>2$
2501.06662_F00123	007	—	$P_n=P_1$	$P_n=P_1$
2501.06662_F00124	007	—	$a'\in A$	$a'\in A$
2501.06662_F00125	007	—	$P_n(a,\dagger)=0$	$P_n(a,\dagger)=0$
2501.06662_F00126	007	—	$P_n _{A\times A}$	$P_n _{A\times A}$
2501.06662_F00127	007	—	$(P_n)_{n\geq 1}$	$(P_n)_{n\geq 1}$
2501.06662_F00128	007	—	$q:A\rightarrow[0,1]$	$q:A\rightarrow[0,1]$
2501.06662_F00129	007	—	$\sum_{a'\in A}q(a)P(a,a')=q(a')$	$\sum_{a'\in A}q(a)P(a,a')=q(a')$
2501.06662_F00130	007	—	$p_{\perp}(a)=q(a)$	$p_{\perp}(a)=q(a)$
2501.06662_F00131	008	—	$n-1$	$n-1$
2501.06662_F00132	008	—	$p_{\perp a_1\cdots a_i}$	$p_{\perp a_1\cdots a_i}$
2501.06662_F00133	008	—	$\perp a_1\cdots a_i$	$\perp a_1\cdots a_i$
2501.06662_F00134	008	—	$i<n-2$	$i<n-2$
2501.06662_F00135	008	—	$p^{(\theta)}:A^*\rightarrow\Delta(A),\quad a\mapsto p_{\perp a}^{(\theta)}(-)$	$p^{(\theta)}:A^*\rightarrow\Delta(A),\quad a\mapsto p_{\perp a}^{(\theta)}(-)$
2501.06662_F00136	008	—	$\Delta(A)$	$\Delta(A)$
2501.06662_F00137	008	—	θ	θ
2501.06662_F00138	008	—	$(a_1,\ldots,a_{n-1})$	$(a_1,\ldots,a_{n-1})$
2501.06662_F00139	008	—	$(a_1,\ldots,a_n)$	$(a_1,\ldots,a_n)$
2501.06662_F00140	008	—	$C\subset A^*$	$C\subset A^*$
2501.06662_F00141	008	—	$y=x a'$	$y=x a'$
2501.06662_F00142	008	—	$ a' =N-\mid x\mid$	$ a' =N-\mid x\mid$
2501.06662_F00143	008	—	aa'	aa'
2501.06662_F00144	008	—	$y=x a''\dagger$	$y=x a''\dagger$
2501.06662_F00145	008	—	$a''\in A^*$	$a''\in A^*$
2501.06662_F00146	008	—	$ a'' \leq N-\mid x\mid-1$	$ a'' \leq N-\mid x\mid-1$
2501.06662_F00147	008	—	aa''	aa''
2501.06662_F00148	008	—	$N>5$	$N>5$
2501.06662_F00149	008	—	$b=$	$b=$
2501.06662_F00150	008	—	$a_1 a_2 a_3 a_4 a_5\in A^*$	$a_1 a_2 a_3 a_4 a_5\in A^*$
2501.06662_F00151	008	—	$a=a_1 a_2$	$a=a_1 a_2$
2501.06662_F00152		—	$f(x)$	$f(x)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value				
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00153		—	f(x) \leq N-1	f(x) ≤ N − 1
2501.06662_F00154		—	p_{f(x)}	p f(x)
2501.06662_F00155	009	—	x=\perp a_{\{1\}} \cdots a_{\{t\}}	x = ⊥ a ₁ ⋯ a _t
2501.06662_F00156	009	—	y=x a_{\{t+1\}} \cdots a_{\{t+k\}}	y = x a _{t+1} ⋯ a _{t+k}
2501.06662_F00157	009	—	k \geq 1	k ≥ 1
2501.06662_F00158	009	—	\left(a_i\right)_{i=1}^{t+k-1} \subset A	(a _i) _{i=1} ^{t+k−1} ⊂ A
2501.06662_F00159	009	—	a_{\{t+k\}} \in A \cup \{\dagger\}	a _{t+k} ∈ A ∪ {†}
2501.06662_F00160	009	—	y_{<t+i}	y<t+i
2501.06662_F00161	009	—	\perp a_{\{1\}} \cdots a_{\{t+i-1\}}	⊥ a ₁ ⋯ a _{t+i−1}
2501.06662_F00162	009	—	y_{<t+1}=x	y<t+1 = x
2501.06662_F00163	009	—	m=0	m = 0
2501.06662_F00164	009	—	T(x)=\{x\}	T(x) = {x}
2501.06662_F00165	009	—	m=1	m = 1
2501.06662_F00166	009	—	T(x)=\{x~a~ ~a~\in~A~\cup~\{\dagger\}\}	T(x) = {xa a ∈ A ∪ {†}}
2501.06662_F00167	009	—	m \geq 1	m ≥ 1
2501.06662_F00168	009	—	a \in A^m	a ∈ A ^m
2501.06662_F00169	009	—	\left(a^\prime,a''\right)\in A^{m-1}\times A	(a′,a″) ∈ A ^{m−1} × A
2501.06662_F00170	009	—	i=m-1	i = m − 1
2501.06662_F00171	009	—	\pi\!\left(x~a^\prime~ ~x\right)p\!\left(\dagger~ ~xa'\right)	π(xa′ x) p(† xa′)
2501.06662_F00172	009	—	a^\prime \in A^{m-1}	a′ ∈ A ^{m−1}
2501.06662_F00173		—	\square	□
2501.06662_F00174	010	—	(\mathcal{V}, \leq, \otimes, 1)	(V, ≤, ⊗, 1)
2501.06662_F00175	010	—	(\mathcal{V}, \leq)	(V, ≤)
2501.06662_F00176	010	—	(\mathcal{V}, \otimes, 1)	(V, ⊗, 1)
2501.06662_F00177	010	—	x \otimes y \leq x^\prime \otimes y^\prime	x ⊗ y ≤ x′ ⊗ y′
2501.06662_F00178	010	—	x \leq x^\prime	x ≤ x′
2501.06662_F00179	010	—	y \leq y^\prime	y ≤ y′
2501.06662_F00180	010	—	([0,1], \leq, \cdot, 1)	([0, 1], ≤, ·, 1)
2501.06662_F00181	010	—	a~b:=a\cdot b	ab := a · b
2501.06662_F00182	010	—	a,\,b\in [0,1]	a, b ∈ [0, 1]
2501.06662_F00183	010	—	[0,\infty], \geq, +, 0	[0, ∞], ≥, +, 0
2501.06662_F00184	010	—	a+\infty:=\infty	a + ∞ := ∞
2501.06662_F00185	010	—	\infty+a:=\infty	∞ + a := ∞
2501.06662_F00186	010	—	a \in [0,\infty]	a ∈ [0, ∞]
2501.06662_F00187	010	—	\mathcal{V}	V
2501.06662_F00188	010	—	\mathcal{C}	C
2501.06662_F00189	010	—	\operatorname{ob}(\mathcal{C})	ob(C)
2501.06662_F00190	010	—	\mathcal{C}(x,y)	C(x, y)
2501.06662_F00191	010	—	x,y,z\in \operatorname{ob}(\mathcal{C})	x, y, z ∈ ob(C)
2501.06662_F00192	010	—	\pi(y\mid x)=0	π(y x) = 0
2501.06662_F00193	010	—	x=y	x = y
2501.06662_F00194	010	—	\mathcal{L}(x,y):=\pi(y\mid x)	L(x, y) := π(y x)
2501.06662_F00195	011	—	x,\,y,\,z	x, y, z
2501.06662_F00196	011	—	x \rightarrowtail y \rightarrowtail z	x → y → z
2501.06662_F00197	011	—	x \neq y	x ≠ y
2501.06662_F00198	011	—	y \neq z	y ≠ z
2501.06662_F00199	011	—	\left(a_i\right)_{i=1}^{t+k+k^\prime-1} \subset A	(a _i) _{i=1} ^{t+k+k′−1} ⊂ A
2501.06662_F00200	011	—	a_{\{t+k+k^\prime\}} \in A \cup \{\dagger\}	a _{t+k+k′} ∈ A ∪ {†}
2501.06662_F00201	011	—	\pi(y\mid x)=1	π(y x) = 1
2501.06662_F00202	011	—	\pi(z\mid y)=\pi(z\mid x)	π(z y) = π(z x)
2501.06662_F00203	011	—	z=y	z = y
2501.06662_F00204	011	—	z	z
2501.06662_F00205	011	—	\pi(y\mid x)\,\pi(z\mid y)\leq\pi(z\mid x)	π(y x)π(z y) ≤ π(z x)
2501.06662_F00206	011	—	\pi(y\mid x),\,\pi(z\mid x)\not=0	π(y x), π(z x) ≠ 0
2501.06662_F00207	011	—	\pi(z\mid y)=0,{ }^3	π(z y) = 0, ³
2501.06662_F00208	011	—	\pi(z\mid y),\,\pi(z\mid x)\not=0	π(z y), π(z x) ≠ 0
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value				
Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00209	011	—	$\pi(y \mid x)=0 \text{ .}\{ \}^{\wedge\{4\}}$	$\pi(y \mid x) = 0.$ ⁴
2501.06662_F00210	011	—	$[0,1]$	$[0,1]$
2501.06662_F00211	011	—	$\mathcal{L}(x,y)$	$\mathcal{L}(x,y)$
2501.06662_F00212		—	$x=\perp$	$x=\perp$
2501.06662_F00213		—	$y=\perp$	$y=\perp$
2501.06662_F00214		—	$z=\perp$	$z=\perp$
2501.06662_F00215	012	—	$d(x,y)+d(y,z) \geq d(x,z)$	$d(x,y)+d(y,z) \geq d(x,z)$
2501.06662_F00216	012	—	$d(x,x)=0$	$d(x,x)=0$
2501.06662_F00217	012	—	∞	∞
2501.06662_F00218	012	—	$\mathcal{M}(x,y):=d(x,y)$	$\mathcal{M}(x,y):=d(x,y)$
2501.06662_F00219	012	—	$\mathrm{L}^{\leq N}$	$\mathrm{L}^{\leq N}$
2501.06662_F00220	012	—	R	R
2501.06662_F00221	012	—	$\mid\!-\!\mid\!:\operatorname{ob}(\mathcal{V})\rightarrow R$	$\mid\!-\!\mid\!:\operatorname{ob}(\mathcal{V})\rightarrow R$
2501.06662_F00222	012	—	$\mid\!v\!\mid\!=\!\mid\!w\!\mid\!$	$\ v\ =\ w\ $
2501.06662_F00223	012	—	$v\cong w$	$v\cong w$
2501.06662_F00224	012	—	$\mid\!1\!\mid\!=1$	$\ 1\ =1$
2501.06662_F00225	012	—	$\mid\!v\otimes w\!\mid\!=\!\mid\!v\!\mid\!\mid\!w\!\mid\!$	$\ v\otimes w\ =\ v\ \ w\ $
2501.06662_F00226	012	—	$v,w\in\operatorname{ob}(\mathcal{V})$	$v,w\in\operatorname{ob}(\mathcal{V})$
2501.06662_F00227	012	—	$\zeta_{\mathcal{C}}\colon\operatorname{ob}(\mathcal{C})\times\operatorname{ob}(\mathcal{C})\rightarrow R$	$\zeta_{\mathcal{C}}:\operatorname{ob}(\mathcal{C})\times\operatorname{ob}(\mathcal{C})\rightarrow R$
2501.06662_F00228	012	—	$\zeta_{\mathcal{C}}(x,y):=\mid\!\mathcal{C}(x,y)\!\mid\!$	$\zeta_{\mathcal{C}}(x,y):=\ \mathcal{C}(x,y)\ $
2501.06662_F00229	012	—	$\zeta_{\mathcal{C}}$	$\zeta_{\mathcal{C}}$
2501.06662_F00230	012	—	$\mu_{\mathcal{C}}:=\zeta_{\mathcal{C}}^{-1}$	$\mu_{\mathcal{C}}:=\zeta_{\mathcal{C}}^{-1}$
2501.06662_F00231	012	—	$\operatorname{Mag}(\mathcal{C})$	$\operatorname{Mag}(\mathcal{C})$
2501.06662_F00232	013	—	$\mathcal{V}=[0,\infty]$	$\mathcal{V}=[0,\infty]$
2501.06662_F00233	013	—	$\mid\!x\!\mid\!_t=e^{-t\,x}$	$\ x\ _t=e^{-tx}$
2501.06662_F00234	013	—	$t\in\mathbb{R}\cup\{\infty\}$	$t\in\mathbb{R}\cup\{\infty\}$
2501.06662_F00235	013	—	$\mid\!-\!\mid\!_1$	$\ -\ _1$
2501.06662_F00236	013	—	$f(t):=\operatorname{Mag}(t\mathcal{C})$	$f(t):=\operatorname{Mag}(t\mathcal{C})$
2501.06662_F00237	013	—	$t\in(0,\infty)$	$t\in(0,\infty)$
2501.06662_F00238	013	—	$t\mathcal{C}$	$t\mathcal{C}$
2501.06662_F00239	013	—	$[s,t]=\{u\in P:s\leq u\leq t\}$	$[s,t]=\{u\in P:s\leq u\leq t\}$
2501.06662_F00240	013	—	$s\leq t$	$s\leq t$
2501.06662_F00241	013	—	$\mathbb{k}$	$\mathbb{k}$
2501.06662_F00242	013	—	$\operatorname{int}(P)$	$\operatorname{int}(P)$
2501.06662_F00243	013	—	$I(P,\mathbb{k})$	$I(P,\mathbb{k})$
2501.06662_F00244	013	—	$f\colon\operatorname{int}(P)\rightarrow\mathbb{k}$	$f:\operatorname{int}(P)\rightarrow\mathbb{k}$
2501.06662_F00245	013	—	$f(s,t)$	$f(s,t)$
2501.06662_F00246	013	—	$f([s,t])$	$f([s,t])$
2501.06662_F00247	013	—	$\delta\colon\operatorname{int}(P)\rightarrow\mathbb{k}$	$\delta:\operatorname{int}(P)\rightarrow\mathbb{k}$
2501.06662_F00248	013	—	$f*\delta=\delta*f=f$	$f*\delta=\delta*f=f$
2501.06662_F00249	013	—	$f\in I(P,\mathbb{k})$	$f\in I(P,\mathbb{k})$
2501.06662_F00250	013	—	ζ_P	ζ_P
2501.06662_F00251	013	—	$\mathbf{2}:=\{$	$\mathbf{2}:=\{$
2501.06662_F00252	013	—	$\}$	$\}$
2501.06662_F00253	013	—	$s,u\in P$	$s,u\in P$
2501.06662_F00254	013	—	$s\leq u$	$s\leq u$
2501.06662_F00255	013	—	$P(s,u)$	$P(s,u)$
2501.06662_F00256	013	—	$s\not\leq u$	$s\not\leq u$
2501.06662_F00257	013	—	$\mid\!-\!\mid\!:\operatorname{ob}(\mathbf{2})\rightarrow\mathbb{Z}$	$\mid\!-\!\mid\!:\operatorname{ob}(\mathbf{2})\rightarrow\mathbb{Z}$
2501.06662_F00258	013	—	$\mid\!$	$\ $
2501.06662_F00259	013	—	$\mid\!=1$	$\ = 1$
2501.06662_F00260	013	—	$\mid\!=0$	$\ = 0$
2501.06662_F00261	013	—	$\zeta_P(s,u)=\mid\!P(s,u)\!\mid\!$	$\zeta_P(s,u)=\ P(s,u)\ $
2501.06662_F00262	013	—	μ_P	μ_P
2501.06662_F00263	014	—	μ_{L}	μ_{L}

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00264	014	—	$x \in \operatorname{ob}(\operatorname{\mathsf{L}})$	$x \in \mathbf{ob}(\operatorname{\mathsf{L}})$
2501.06662_F00265	014	—	$a_{\{1\}}, a_{\{2\}}, \ldots \in A \cup \{\dagger\}$	$a_1, a_2, \ldots \in A \cup \{\dagger\}$
2501.06662_F00266	014	—	$\mu_{\operatorname{\mathsf{L}}}(\left(x, x_{a_{\{1\}}} a_{\{2\}}} \cdots a_{\{j\}}\right)=0$	$\mu_{\operatorname{\mathsf{L}}}(x, xa_1a_2\cdots a_j)=0$
2501.06662_F00267	014	—	$j>1$	$j>1$
2501.06662_F00268	014	—	$\# A$	$\#A$
2501.06662_F00269	014	—	$\#$	$\#$
2501.06662_F00270	014	—	$(\operatorname{\mathsf{L}})$	$(\operatorname{\mathsf{L}})$
2501.06662_F00271	015	—	$\operatorname{\mathsf{L}}^{\leq N}=\bigsqcup_{i=0}^{N-1}\operatorname{\mathsf{L}}^{\{i\}}$	$\operatorname{\mathsf{L}}^{\leq N}=\bigsqcup_{i=0}^{N-1}\operatorname{\mathsf{L}}^{(i)}$
2501.06662_F00272	015	—	$\operatorname{\mathsf{L}}^{\{0\}}=\{\perp\}$	$\operatorname{\mathsf{L}}^{(0)}=\{\perp\}$
2501.06662_F00273	015	—	$i\leq 1, \operatorname{\mathsf{L}}^{\{i\}}$	$i\leq 1, \operatorname{\mathsf{L}}^{(i)}$
2501.06662_F00274	015	—	$a\in A^{\dagger}$	$a\in A^{\dagger}$
2501.06662_F00275	015	—	$\perp a^{\prime \dagger}$	$\perp a^{\prime \dagger}$
2501.06662_F00276	015	—	$a^{\prime }\in A^{i-1}$	$a^{\prime }\in A^{i-1}$
2501.06662_F00277	015	—	$\#\operatorname{\mathsf{L}}^{\{i\}}=\#A^{\{i\}}+\#A^{\{i-1\}}$	$\#\operatorname{\mathsf{L}}^{(i)}=\#A^i+\#A^{i-1}$
2501.06662_F00278	015	—	$\operatorname{Mag}(\operatorname{\mathsf{L}})=1$	$\mathbf{Mag}(\operatorname{\mathsf{L}})=1$
2501.06662_F00279	015	—	$\operatorname{ob}(\operatorname{\mathsf{M}})=\operatorname{ob}(\left(\operatorname{\mathsf{L}}^{\leq N}\right))$	$\mathbf{ob}(\mathcal{M})=\mathbf{ob}\left(\operatorname{\mathsf{L}}^{\leq N}\right)$
2501.06662_F00280	015	—	$\operatorname{\mathsf{M}}(x,y)=d(x,y)=-\ln \pi (y\mid x)$	$\mathcal{M}(x,y)=d(x,y)=-\ln \pi (y\mid x)$
2501.06662_F00281	015	—	$\zeta _{\operatorname{\mathsf{M}}}: \operatorname{ob}(\operatorname{\mathsf{M}})\times \operatorname{ob}(\operatorname{\mathsf{M}})\rightarrow \mathbb{R}$	$\zeta _{\mathcal{M}}:\mathbf{ob}(\mathcal{M})\times \mathbf{ob}(\mathcal{M})\rightarrow \mathbb{R}$
2501.06662_F00282	015	—	$d(x,y)=\infty$	$d(x,y)=\infty$
2501.06662_F00283	015	—	$\zeta _t$	$\zeta _t$
2501.06662_F00284	015	—	$(\zeta _{\operatorname{\mathsf{M}}})_t$	$(\zeta _{\mathcal{M}})_t$
2501.06662_F00285	015	—	$\zeta _t^{-1}$	$\zeta _t^{-1}$
2501.06662_F00286	015	—	$\operatorname{ob}(\operatorname{\mathsf{M}})$	$\mathbf{ob}(\mathcal{M})$
2501.06662_F00287	015	—	$x=y_{\{0\}}\rightarrow y_{\{1\}}\rightarrow \cdots \rightarrow y_k=y$	$x=y_0\rightarrow y_1\rightarrow \cdots \rightarrow y_k=y$
2501.06662_F00288	016	—	δ	δ
2501.06662_F00289	016	—	$\delta (x,y)=1$	$\delta (x,y)=1$
2501.06662_F00290	016	—	$\delta (x,y)=0$	$\delta (x,y)=0$
2501.06662_F00291	016	—	$\zeta -\delta$	$\zeta -\delta$
2501.06662_F00292	016	—	D	D
2501.06662_F00293	016	—	$E=\left\{\left(z,z^{\prime }\right)\mid z\rightarrow z^{\prime }\right\}$	$E=\left\{\left(z,z^{\prime }\right)\mid z\rightarrow z^{\prime }\right\}$
2501.06662_F00294	016	—	$\left.\operatorname{\mathsf{L}},z\neq z^{\prime }\right\}$	$\left.\operatorname{\mathsf{L}},z\neq z^{\prime }\right\}$
2501.06662_F00295	016	—	$(\zeta _t-\delta)^k$	$(\zeta _t-\delta)^k$
2501.06662_F00296	016	—	k	k
2501.06662_F00297	016	—	E	E
2501.06662_F00298	016	—	(x,y)	(x,y)
2501.06662_F00299	016	—	$\zeta _{\operatorname{\mathsf{L}}}^{-1}$	$\zeta _{\operatorname{\mathsf{L}}}^{-1}$
2501.06662_F00300	016	—	$\zeta _{\operatorname{\mathsf{L}}}$	$\zeta _{\operatorname{\mathsf{L}}}$
2501.06662_F00301	017	—	$k\geq 0$	$k\geq 0$
2501.06662_F00302	017	—	$\zeta _t^{-1}(x,y)$	$\zeta _t^{-1}(x,y)$
2501.06662_F00303	017	—	$\pi (y\mid x)^t\zeta _{\operatorname{\mathsf{L}}}^{-1}(x,y)$	$\pi (y\mid x)^t\zeta _{\operatorname{\mathsf{L}}}^{-1}(x,y)$
2501.06662_F00304	017	—	$\zeta _{\operatorname{\mathsf{L}}}^{-1}(x,y)=\mu _{\operatorname{\mathsf{L}}}(x,y)$	$\zeta _{\operatorname{\mathsf{L}}}^{-1}(x,y)=\mu _{\operatorname{\mathsf{L}}}(x,y)$
2501.06662_F00305	017	—	$f:(0,\infty)\rightarrow \mathbb{R}$	$f:(0,\infty)\rightarrow \mathbb{R}$
2501.06662_F00306	017	—	$f(t)=\operatorname{Mag}(t\operatorname{\mathsf{M}})$	$f(t)=\mathbf{Mag}(t\mathcal{M})$
2501.06662_F00307	017	—	$p=(p_1,\ldots ,p_n)$	$p=(p_1,\ldots ,p_n)$
2501.06662_F00308	017	—	$(p_x:A\cup \{\dagger \}\rightarrow [0,1])_{x\in \operatorname{ob}(\operatorname{\mathsf{L}})}$	$(p_x:A\cup \{\dagger \}\rightarrow [0,1])_{x\in \mathbf{ob}(\operatorname{\mathsf{L}})}$
2501.06662_F00309	017	—	$T(\perp)$	$T(\perp)$
2501.06662_F00310		—	$\operatorname{Mag}(t\operatorname{\mathsf{M}})=\sum _{x,y\in \operatorname{ob}(\operatorname{\mathsf{M}})}\pi (y\mid x)^t\zeta _{\operatorname{\mathsf{L}}}^{-1}(x,y)$	$\mathbf{Mag}(t\mathcal{M})=\sum _{x,y\in \mathbf{ob}(\mathcal{M})}\pi (y\mid x)^t\zeta _{\operatorname{\mathsf{L}}}^{-1}(x,y)$
2501.06662_F00311	018	—	$y\neq x$	$y\neq x$
2501.06662_F00312	018	—	$\operatorname{ob}(\operatorname{\mathsf{M}})\backslash T(\perp)$	$\mathbf{ob}(\mathcal{M})\backslash T(\perp)$
2501.06662_F00313	018	—	$\operatorname{ob}(\operatorname{\mathsf{M}})=T(\perp)\sqcup (\operatorname{ob}(\operatorname{\mathsf{M}})\backslash T(\perp))$	$\mathbf{ob}(\mathcal{M})=T(\perp)\sqcup (\mathbf{ob}(\mathcal{M})\backslash T(\perp))$
2501.06662_F00314	018	—	$\zeta _t^{-1}(x,x)=1$	$\zeta _t^{-1}(x,x)=1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00315	018	—	$x \in \operatorname{ob}(\mathcal{M})$	$x \in \mathbf{ob}(\mathcal{M})$
2501.06662_F00316	018	—	$t \neq 1$	$t \neq 1$
2501.06662_F00317	018	—	$(t-1)/(t-1)$	$(t-1)/(t-1)$
2501.06662_F00318	018	—	$t \geq 1$	$t \geq 1$
2501.06662_F00319	018	—	$p: S \rightarrow [0,1]$	$p: S \rightarrow [0,1]$
2501.06662_F00320	018	—	S	S
2501.06662_F00321	018	—	$H(p)$	$H(p)$
2501.06662_F00322	018	—	p	p
2501.06662_F00323	019	—	$E_{\{x\}}(a)=$	$E_x(a)=$
2501.06662_F00324	019	—	$-\ln p_{\{x\}}(a)=d(x, \, x \, a)$	$-\ln p_x(a)=d(x, xa)$
2501.06662_F00325	019	—	$f(t)=$	$f(t)=$
2501.06662_F00326	019	—	$\#(\operatorname{ob}(\mathcal{M}))-\tilde{Z}(t)$	$\#(\mathbf{ob}(\mathcal{M}))-\tilde{Z}(t)$
2501.06662_F00327	019	—	$\tilde{Z}$	$\tilde{Z}$
2501.06662_F00328	019	—	$(x, \, a) \in$	$(x, a) \in$
2501.06662_F00329	019	—	$S:=(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) \times (A \cup \{\dagger\})$	$S:=(\mathbf{ob}(\mathcal{M}) \setminus T(\perp)) \times (A \cup \{\dagger\})$
2501.06662_F00330	019	—	$E(x, \, a)=d(x, \, a)=-\ln p_{\{x\}}(a)$	$E(x, a)=d(x, a)=-\ln p_x(a)$
2501.06662_F00331	019	—	ρ	ρ
2501.06662_F00332	019	—	$\rho(s)=$	$\rho(s)=$
2501.06662_F00333	019	—	$e^{-t \, E(s)} / \tilde{Z}(t)$	$e^{-tE(s)} / \tilde{Z}(t)$
2501.06662_F00334	019	—	$y, \, z \in \operatorname{ob}\nolimits\!\left(\mathcal{M}_{\{x\}}\right)$	$y, z \in \mathbf{ob}\nolimits\!\left(\mathcal{M}_x\right)$
2501.06662_F00335	019	—	$\mathcal{M}_{\{x\}}(y, \, z):=\mathcal{M}(y, \, z)=-\ln \pi(z \, \mid y)$	$\mathcal{M}_x(y, z):=\mathcal{M}(y, z)=-\ln \pi(z \mid y)$
2501.06662_F00336	020	—	$M, \, d$	M, d
2501.06662_F00337	020	—	$\ell \in [0, \, \infty)$	$\ell \in [0, \infty)$
2501.06662_F00338	020	—	$\left(M \, C_{\{k, \, \ell\}}(M)\right)_{k \in \mathbb{N}}$	$(MC_{k,\ell}(M))_{k \in \mathbb{N}}$
2501.06662_F00339	020	—	$M \, C_{\{k, \, \ell\}}(M):=\mathbb{Z}\left[G_{\{k, \, \ell\}}\right]$	$MC_{k,\ell}(M):=\mathbb{Z}[G_{k,\ell}]$
2501.06662_F00340	020	—	$\partial_k: M \, C_{\{k, \, \ell\}}(M) \rightarrow M \, C_{\{k-1, \, \ell\}}(M)$	$\partial_k: MC_{k,\ell}(M) \rightarrow MC_{k-1,\ell}(M)$
2501.06662_F00341	020	—	$0< i < k$	$0 < i < k$
2501.06662_F00342	020	—	$\partial_{\{k\}}^0=\partial_{\{k\}}^k=0$	$\partial_k^0=\partial_k^k=0$
2501.06662_F00343	020	—	$[0, \, \infty)$	$[0, \infty)$
2501.06662_F00344	020	—	$\left(M \, C_{\{\bullet, \, \ell\}}(M), \, \partial_{\bullet}\right)$	$(MC_{\bullet,\ell}(M), \partial_{\bullet})$
2501.06662_F00345	020	—	M	M
2501.06662_F00346	020	—	$H_{\{\bullet \, \bullet\}}(M)$	$H_{\bullet\bullet}(M)$
2501.06662_F00347	020	—	$\left(M \, C_{\{\bullet\}}, \, \ell_{\ell}\right)_{\ell}$	$(MC_{\bullet,\ell})_{\ell}$
2501.06662_F00348	021	—	$q=e^{-t}$	$q=e^{-t}$
2501.06662_F00349	021	—	$\bigcup_{k \geq 0} M \, C_{\{k, \, \ell\}}(\mathcal{M}) \neq$	$\bigcup_{k \geq 0} MC_{k,\ell}(\mathcal{M}) \neq$
2501.06662_F00350	021	—	$\emptyset$	$\emptyset$
2501.06662_F00351	021	—	$x=y_{\{0\}} \rightarrow$	$x=y_0 \rightarrow$
2501.06662_F00352	021	—	$y_{\{1\}} \rightarrow \cdots \rightarrow y_{\{k\}}=y$	$y_1 \rightarrow \cdots \rightarrow y_k=y$
2501.06662_F00353	021	—	$\prod_{i=1}^k \pi\left(y_{\{i\}} \, \mid y_{\{i-1\}}\right) \neq 0$	$\prod_{i=1}^k \pi\left(y_i \mid y_{i-1}\right) \neq 0$
2501.06662_F00354	021	—	$d\left(y_{\{i\}}, \, y_{\{i+1\}}\right)<\infty$	$d\left(y_i, y_{i+1}\right)<\infty$
2501.06662_F00355	021	—	$i=0, \, \ldots, \, k-1$	$i=0, \ldots, k-1$
2501.06662_F00356	021	—	$\ell:=\sum_{i=0}^{k-1} d\left(y_{\{i\}}, \, y_{\{i+1\}}\right)<\infty$	$\ell:=\sum_{i=0}^{k-1} d\left(y_i, y_{i+1}\right)<\infty$
2501.06662_F00357	021	—	$\left(y_{\{0\}}, \, \ldots, \, y_{\{k\}}\right)$	$(y_0, \ldots, y_k)$
2501.06662_F00358	021	—	$G_{\{k, \, \ell\}}$	$G_{k,\ell}$
2501.06662_F00359	021	—	$\prod_{i=1}^k \pi\left(y_{\{i\}} \, \mid y_{\{i-1\}}\right)^t=\exp \left(-t \, \ell\right)$	$\prod_{i=1}^k \pi\left(y_i \mid y_{i-1}\right)^t=\exp(-t\ell)$
2501.06662_F00360	021	—	$M \, C_{\{k, \, \ell\}}$	$MC_{k,\ell}$
2501.06662_F00361	021	—	$\left(M \, C_{\{\bullet\}}, \, \ell\right)_{\ell}$	$(MC_{\bullet,\ell})_{\ell}$
2501.06662_F00362	022	—	$H_{\{0,0\}}(\mathcal{M})$	$H_{0,0}(\mathcal{M})$
2501.06662_F00363	022	—	$\ell>0$	$\ell > 0$
2501.06662_F00364	022	—	$H_{\{0, \, \ell\}}(\mathcal{M})=0$	$H_{0,\ell}(\mathcal{M})=0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00365	022	—	$H_{\{1, \ell\}}(\mathcal{M})$	$H_{1,\ell}(\mathcal{M})$
2501.06662_F00366	022	—	$d(x, y)=\ell$	$d(x,y)=\ell$
2501.06662_F00367	022	—	$y=a_{\{0\}}\,a_{\{1\}}\,\cdots\,a_{\{n\}}$	$y=a_0a_1\cdots a_n$
2501.06662_F00368	022	—	$p\left(-\mid y_{\{<i\}}\right)$	$p(-\mid y_{<i})$
2501.06662_F00369	022	—	$y_{\{<i\}}$	$y_{<i}$
2501.06662_F00370	022	—	$\zeta_t\!\left(a_{\{0\}},\,y\right)$	$\zeta_t(a_0,y)$
2501.06662_F00371	022	—	$t=1\ /\ n$	$t=1/n$
2501.06662_F00372	022	—	$a_{\{0\}}$	a_0
2501.06662_F00373	022	—	$\zeta_t\!\left(a_{\{0\}},\,y\right)=1\ /\ P\,P\,L(y)$	$\zeta_t(a_0,y)=1/PPL(y)$
2501.06662_F00374	022	—	$\zeta_t\!\left(x,\,y\right)$	$\zeta_t(x,y)$
2501.06662_F00375	022	—	$\widehat{\mathcal{M}}:=[0,\infty)^{\mathcal{M}}$	$\widehat{\mathcal{M}}:=[0,\infty]^{\mathcal{M}}$
2501.06662_F00376	022	—	$f:\mathcal{M}\rightarrow[0,\infty]$	$f:\mathcal{M}\rightarrow[0,\infty]$
2501.06662_F00377	022	—	$\max\left\{f(y)-f(x),\,0\right\}\leq\mathcal{M}(x,y)$	$\max\{f(y)-f(x),0\}\leq\mathcal{M}(x,y)$
2501.06662_F00378	022	—	$\widehat{\mathcal{M}}$	$\widehat{\mathcal{M}}$
2501.06662_F00379	023	—	$p:\operatorname{ob}(\mathrm{A})\rightarrow[0,1]$	$p:\mathbf{ob}(\mathbf{A})\rightarrow[0,1]$
2501.06662_F00380	023	—	$\theta:\operatorname{ob}(\mathrm{A})\times\operatorname{ob}(\mathrm{A})\rightarrow[0,\infty)$	$\theta:\mathbf{ob}(\mathbf{A})\times\mathbf{ob}(\mathbf{A})\rightarrow[0,\infty)$
2501.06662_F00381	023	—	$\theta=\zeta_t$	$\theta=\zeta_t$
2501.06662_F00382	023	—	$p=\left.\pi(-\mid x)\right _{T(x)}$	$p=\pi(-\mid x) _{T(x)}$
2501.06662_F00383	023	—	$\left.\pi(-\mid x)\right _{T(x)}$	$\pi(-\mid x) _{T(x)}$
2501.06662_F00384	023	—	$P:A\times A\rightarrow[0,1]$	$P:A\times A\rightarrow[0,1]$
2501.06662_F00385	023	—	$p_{\{\perp\}}$	$p_{\perp}$
2501.06662_F00386	024	—	$\{ \ }^{\sim}\!\sim$	$\sim$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				